

Lingam Hemang

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Professional Summary

Senior Data Engineer who has **architected and owned production distributed data systems** for Apple Inc. over 4 years at Infosys – making independent design calls across **Spark, Kafka, Airflow, and AWS** that hold **99.5% uptime** with zero data-loss incidents. Those decisions on retry strategy, schema validation, and query optimisation delivered **40% faster pipelines** and a **35% drop in error rate**, on infrastructure I remain accountable for end-to-end. Deep expertise in **ETL engineering, distributed processing, Snowflake, and data modelling**; now extending that ownership into **GenAI applications** using LangChain, RAG pipelines, and AWS Bedrock.

Technical Skills

Languages: Python, SQL, PySpark

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Data Quality, Partitioning & Query Optimisation, Semantic Layer & Materialisation, Tableau

Data Formats: JSON, Parquet, Avro, CSV

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Prompt Engineering

Backend & APIs: Django, Flask, REST API Design & Integration, OOP, SOLID Principles, Design Patterns, pytest

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, Splunk, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Senior Data Engineer

July 2022 – Present

Client: Apple Inc. | Python, PySpark, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, Splunk

- Architected production **ETL pipelines** end-to-end on a large-scale distributed platform (**Spark, Kafka, Airflow, AWS**) – chose distributed processing and SQL query optimisation over naive batch loads to ingest **REST API** data into **Snowflake**, cutting processing time **40%**.
- Designed and ran **high-availability, fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka** via **Airflow DAGs**; engineered retry logic and self-healing schema validation from scratch – sustaining **99.5% uptime** and cutting error rate **35%** with zero data-loss incidents.
- Owned the **semantic layer** from design to production: built materialisation logic in Python, deciding which tables/views to pre-aggregate to balance freshness against **Tableau** query performance; identified a recurring onboarding bottleneck and built **reusable framework extensions**, cutting onboarding effort **30%**.
- Extended the **Splunk connector** with custom queries to load usage data into Snowflake and designed a **Tableau security audit dashboard** from the ground up to surface unauthorised access patterns proactively.
- Owned **data consistency** for a production application – designed the validation logic behind database views and the load process into a **semantic search cluster**; managed releases via **CI/CD** pipelines and set data governance standards across Agile feature teams.

Projects

Agentic AI Git Diff Analyzer | *Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab* | [Repo](#)

- Designed a **dual-mode SQL schema impact analyzer** – local (Ollama) and **AWS mode** (Bedrock Claude 3.5 via GitHub Webhook → Lambda → EC2); built downstream dependency detection across the repo and **LLM impact reports** with risk-level badges as styled PDFs; real-time **Django UI** with **SSE** streaming.

Document Intelligence Chatbot – RAG Pipeline | *Python, LangChain, ChromaDB, Django, Ollama, EasyOCR* | [Repo](#)

- Designed an **end-to-end RAG pipeline** (LangChain + ChromaDB + Ollama) with persistent multi-turn memory; built multi-format ingestion via **PyMuPDF + EasyOCR**, an incremental vector store with snapshot/rollback, and a **10-endpoint Django REST API** with live model switching.

Retail ML Sales Forecasting API | *Python, Flask, scikit-learn, XGBoost* | [Repo](#) **Infosys Hackathon – 2nd Place**

- Built a **multi-model ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales and product recommendations via **Flask REST API**; designed cyclical sin/cos encoding for seasonality; shipped an interactive **Dash** + **Plotly** dashboard.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0